

ServeSID: Hierarchy-Aligned and Direction-Aware Knowledge Distillation for Semantic-ID Generative Recommendation

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Abstract

Generative recommendation with Semantic IDs (SIDs) ranks items by accumulating token-level scores, but autoregressive decoding makes strong teachers costly to serve. Lightweight, verify-free students reduce inference overhead, yet conventional gold-prefix token distillation does not directly supervise preferences over complete SID candidates or their shared-prefix groups. Our diagnostics reveal two structural supervision gaps in conventional gold-prefix token distillation. First, local conditional-distribution agreement does not directly ensure aligned preferences over complete SID candidates. Moreover, teacher and student may assign different probability masses to hierarchical prefix groups. Second, teacher-student relative-score residuals frequently exhibit opposing signs across SID depths, making discrepancy magnitude alone insufficient for identifying directionally favorable corrections. We propose SERVESID to organize teacher supervision over complete SID candidates from two complementary views. HServe aligns complete-candidate distributions and their hierarchically consistent prefix marginals. DA-CDRS decomposes candidate score gaps into depth-wise increments and allocates relative-score distillation weights according to the effects of individual teacher-increment corrections. Both objectives jointly train a lightweight student without changing its verify-free inference architecture. **[TODO: After the no-pair-loss checkpoints are verified, state the overall performance conclusion here, e.g., NDCG@10 across the nine settings and the largest relative gain over base KD.]**

Keywords

Semantic ID; generative recommendation; knowledge distillation; serving; hierarchical alignment

1 Introduction

Generative recommendation represents items as sequences of Semantic-ID (SID) tokens and retrieves them through constrained autoregressive decoding. TIGER [8] ranks complete candidates using accumulated conditional log-probabilities along their SID paths. Although effective, repeated Transformer decoding incurs substantial online cost. Lightweight students such as SID-MLP [2] encode user histories once and predict SID tokens with compact heads, enabling verify-free beam search. Unlike draft-verify alternatives,

these students require no online teacher check, so their recommendation quality must be improved through training rather than inference-time correction. Knowledge distillation transfers teacher preferences to these efficient architectures, but the organization of supervision remains critical to recommendation quality.

Conventional gold-prefix token distillation matches conditional distributions along the observed SID path, whereas recommendation decisions compare complete candidates scored through accumulated token probabilities. Under limited student capacity and training coverage, local alignment does not directly guarantee complete-candidate preference agreement. This raises a central question: how can the generative structure of SIDs be exploited to transfer complete-candidate preferences more effectively without changing the student’s serving architecture?

First, matching local conditional distributions along the observed SID path does not explicitly align teacher and student preferences over complete candidates. This gap motivates complete-candidate distribution matching, particularly under limited student capacity. Beyond item-level preferences, shared SID prefixes organize candidates into nested groups whose probability masses are obtained by aggregating their descendant items. Teacher-student discrepancies may therefore also appear at multiple prefix granularities. Explicit alignment of these hierarchically consistent marginals provides multi-level optimization constraints derived from the same complete-candidate distribution.

Second, the score gap between two complete SID candidates decomposes into depth-wise relative-score increments. Discrepancies between teacher and student increments can have opposing signs and partially cancel. Consequently, a large local error does not necessarily indicate a correction that favors the teacher-preferred ordering. This motivates depth-wise relative-score distillation that evaluates the directional effect of individual teacher-increment corrections instead of allocating supervision solely by discrepancy magnitude.

We propose SERVESID, a structure-aware distillation framework that exploits two complementary views of the same SID candidate preferences. HServe aggregates complete-candidate probabilities into consistent prefix marginals and performs multi-level distribution alignment. DA-CDRS decomposes complete-candidate score relations along SID depths and allocates supervision to selected relative-score increments. Both objectives jointly optimize the same lightweight student using offline teacher knowledge, while preserving its original verify-free inference process.

Because teacher preference is not a relevance oracle, we assess recommendation quality using held-out next-item Recall and NDCG rather than teacher agreement alone. Experiments on Amazon Beauty, Toys, and Video Games with MLP, GRU, and CNN students examine recommendation utility gains over token-distilled baselines. Controlled ablations compare the structural objectives with simpler leaf-only and uniform-weighting alternatives.

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Our contributions are threefold: 1. We identify structural supervision gaps between gold-prefix token matching and complete-candidate preference transfer, supported by candidate-level, hierarchical, and depth-wise diagnostic evidence. 2. We propose SERVESID, which integrates hierarchically consistent candidate-preference alignment with depth-wise relative-score distillation over shared SID candidate scores. 3. We evaluate SERVESID across three datasets and three lightweight student architectures, examining recommendation performance and the contributions of its two structural objectives.

2 Related Work

2.1 Generative Recommendation with Semantic IDs

Generative recommendation casts retrieval as the generation of discrete item identifiers. TIGER [8] popularized Semantic IDs (SIDs) constructed through quantization and retrieved through constrained autoregressive decoding. Subsequent work explores learnable tokenization and improved item representations, including Action-Piece [4], Learnable Item Tokenization [11], and end-to-end SID learning [7, 9]. These developments primarily address how identifiers are constructed and modeled. Other approaches exploit the prefix structure of generation. RIPOR [14], developed for generative information retrieval, introduces prefix-oriented ranking optimization. APAO [13] targets premature beam pruning through adaptive prefix-aware optimization in generative recommendation. LOHRec [12] exploits order and hierarchy in quantized identifiers for lightweight sequential recommendation. These works establish that prefixes influence retrieval, rather than merely encoding item identity. ServeSID instead studies teacher-to-student supervision over complete SID candidates: its hierarchical targets are probability marginals obtained by aggregating candidates with shared prefixes, not direct prefix-survival objectives.

2.2 Knowledge Distillation for Recommendation

Knowledge distillation transfers teacher predictions or preferences to compact models [3]. In recommendation, Ranking Distillation (RD) [10] emphasizes highly ranked teacher items, while Relaxed Ranking Distillation (RRD) [5] transfers relative item orderings through relaxed permutation supervision. RCE-KD [16] revisits cross-entropy distillation for ranking, relating its effectiveness to candidate sampling and using teacher-student-aware selection with adaptive loss combination. These methods show that recommendation distillation can benefit from supervision beyond local output matching, but they do not specifically organize complete SID preferences through shared-prefix marginals and per-depth score increments.

Generation-oriented distillation offers complementary techniques. GKD [1] uses student-generated trajectories for on-policy knowledge transfer, whereas DistiLLM-2 [6] uses contrastive supervision over teacher- and student-generated responses. Both originate in language-model distillation and can be adapted to generative recommendation. More directly, SID-MLP [2] distills autoregressive SID teachers into lightweight position-wise prediction architectures. SmartGR [15] explicitly introduces hierarchy-aware SID distillation and beam-aware ranking distillation, addressing depth-dependent difficulty and prefix pruning. It is therefore inaccurate to characterize hierarchy or ranking as unexplored in SID distillation.

SeRveSID instead organizes supervision over shared complete SID candidates in two complementary ways. HServe aligns complete-candidate preferences and their consistent prefix marginals, while DA-CDRS allocates depth-wise relative-score supervision using candidate-level comparisons. Their contributions are examined through leaf-only, individual-component, and uniform-weight ablations.

2.3 Efficient Serving for Generative Recommendation

Autoregressive generative recommenders repeatedly evaluate decoding networks while expanding beams, making inference cost a central deployment concern. Speculative decoding and draft-verify approaches accelerate generation through additional proposal or checking stages, whereas architectural simplification and alternative decoding strategies reduce per-step overhead. SID-MLP provides a verify-free route by encoding history once and using lightweight SID prediction heads with constrained beam search. ServeSID adopts this serving configuration without claiming a new decoding algorithm: cached teacher candidates, hierarchical marginals, ranking pairs, and depth-wise teacher statistics are used during offline distillation and discarded at deployment. Consequently, its contribution is improving student training under a fixed inference graph, while offline cache construction, storage, and training overhead remain separate costs that should be measured explicitly.

3 Preliminaries

3.1 SID Recommendation and Verify-Free Students

Let $\mathcal{I}$ denote the item catalog and u a recommendation request containing the user's interaction history. Each item $i \in \mathcal{I}$ is assigned a length- L Semantic ID (SID), $z^i = (z_1^i, \dots, z_L^i)$, whose elements belong to a discrete codebook. A fixed autoregressive teacher T and a lightweight student S assign conditional probabilities p_T and q_S , respectively. Their complete-item scores are

$$s_T(i | u) = \sum_{d=1}^L \log p_T(z_d^i | u, z_{<d}^i), \quad s_S(i | u) = \sum_{d=1}^L \log q_S(z_d^i | u, z_{<d}^i). \quad (1)$$

Recommendations are obtained by expanding valid SID prefixes with constrained beam search and ranking completed candidates by these scores. Unlike the Transformer teacher, the verify-free student encodes the history once and uses lightweight prediction

heads for successive SID positions. It requires neither teacher evaluation nor an additional verifier during serving. Both models operate in the same SID space and obey identical prefix-validity constraints. We keep the teacher and SID assignment fixed, so differences among trained students reflect their learned conditional scoring functions rather than changes to identifier construction.

3.2 Gold-Prefix Token Distillation Baseline

We consider a standard student trained with ground-truth SID cross-entropy and token-level knowledge distillation:

$$\begin{aligned}\mathcal{L}_{\text{base}} &= \mathcal{L}_{\text{digit}} + \lambda_{\text{tok}} \mathcal{L}_{\text{tok}}, \\ \mathcal{L}_{\text{digit}} &= - \sum_{d=1}^L \log q_S(z_d^{\text{gt}} | u, z_{<d}^{\text{gt}}), \\ \mathcal{L}_{\text{tok}} &= \sum_{d=1}^L \text{KL}(p_T(\cdot | u, z_{<d}^{\text{gt}}) \| q_S(\cdot | u, z_{<d}^{\text{gt}})).\end{aligned}\quad (2)$$

Here z_d^{gt} is the observed next-item SID and λ_{tok} controls the distillation strength. Importantly, token KD matches the entire next-code distribution at each gold-path prefix, not merely the probability of the correct code. Its direct targets nevertheless remain conditional predictions at visited prefixes. Because alternative complete candidates may follow prefixes absent from the observed path, matching these local distributions does not directly constrain their joint ranking. We use this objective to define the token-distilled baseline; ServeSID retains the ground-truth term but replaces the token-KL term with its structure-aware objectives.

3.3 Two Structural Views of SID Scoring

Hierarchical candidate organization. A prefix p of length d identifies the group

$$\mathcal{C}^{(d)}(p) = \{i \in \mathcal{I} : z_{1:d}^i = p\}. \quad (3)$$

If p' extends p , then $\mathcal{C}^{(d')}(p') \subseteq \mathcal{C}^{(d)}(p)$. Thus, complete candidates form nested groups across SID depths. On a fixed candidate support, the sum of complete-item preference probabilities within a group defines its prefix marginal. These marginals are mutually consistent because each complete SID belongs to exactly one prefix group at any depth. These groups are induced by SID codes and need not coincide with externally defined semantic product categories.

Additive relative-score composition. For two complete candidates i^+ and i^- , define the score gap for $m \in \{T, S\}$ as

$$G_m = s_m(i^+ | u) - s_m(i^- | u) = \sum_{d=1}^L \Delta G_m^{(d)}, \quad (4)$$

where $\Delta G_m^{(d)}$ is the difference between their conditional log-probabilities at position d . Here depth denotes the SID position, not a neural-network layer. Increments before the candidates' first divergent code cancel exactly; later increments may have opposite signs. The sum, rather than any individual increment, determines the final relative ordering. These two properties—aggregation across shared SID prefixes and decomposition along generation depths—provide the structural basis for HServe and DA-CDRS.

4 Diagnosis: Structural Limitations of Token-Level Distillation

We analyze token-distilled MLP, GRU, and CNN students on Amazon Beauty, Toys, and Video Games using 10,000 held-out requests per dataset. The teacher's top-40 valid complete SIDs define a fixed support for comparing teacher and student scores. No held-out teacher outputs enter training.

4.1 Hierarchical Candidate Preference Mismatch

Gold-prefix token KL evaluates next-token agreement on the observed SID path, while complete-candidate KL compares normalized preferences over the shared support. At prefix depth d , we measure hierarchical drift as

$$\text{HDrift}_d(u) = \text{KL}(\pi_{T,40}^{(d)}(\cdot | u) \| \pi_{S,40}^{(d)}(\cdot | u)), \quad (5)$$

where each marginal sums the probabilities of supported candidates sharing a prefix.

Local-to-candidate preference mismatch. Gold-prefix token KD aligns conditional distributions along observed SID paths but does not directly match preferences over complete candidates. Figure 2(a) shows that lower token KL need not imply lower candidate-level KL: on Video Games, GRU has lower token KL than MLP (0.0827 vs. 0.0882) but higher candidate KL (0.1020 vs. 0.0768). This motivates explicit complete-candidate distribution alignment.

Hierarchical prefix-preference mismatch. Shared SID prefixes define nested candidate groups whose probability masses are aggregated from complete candidates (Figure 1). Figure 2(b) shows preferred-prefix disagreement rates of 13.4%–42.8% at depth one and 31.9%–67.7% at depth two. Since exact candidate-distribution matching also determines prefix marginals, hierarchical alignment adds direct multi-granularity optimization constraints rather than independent teacher information.

4.2 Directionally Mixed Cross-Depth Residuals

We examine teacher–student relative-score residuals $r_d = \Delta G_T^{(d)} - \Delta G_S^{(d)}$ on held-out teacher-ordered candidate pairs. Figure 3 shows that 75.4%–86.3% of pairs contain both positive and negative depth-wise residuals, while the largest absolute residual is negative in 47.6%–51.3% of cases. Thus, discrepancy magnitude alone does not identify corrections that favor the teacher-preferred ordering. This motivates direction-aware allocation of depth-wise distillation supervision.

Together, hierarchical candidate-preference mismatches and directionally mixed depth residuals motivate the two complementary structural objectives introduced next.

5 SERVESID: Hierarchy-Aligned and Direction-Aware Distillation

SERVE SID transfers teacher knowledge through two complementary structures of Semantic-ID (SID) scoring. HServe aggregates probabilities of complete candidates into nested prefix groups, while DA-CDRS decomposes complete-candidate ranking relations into

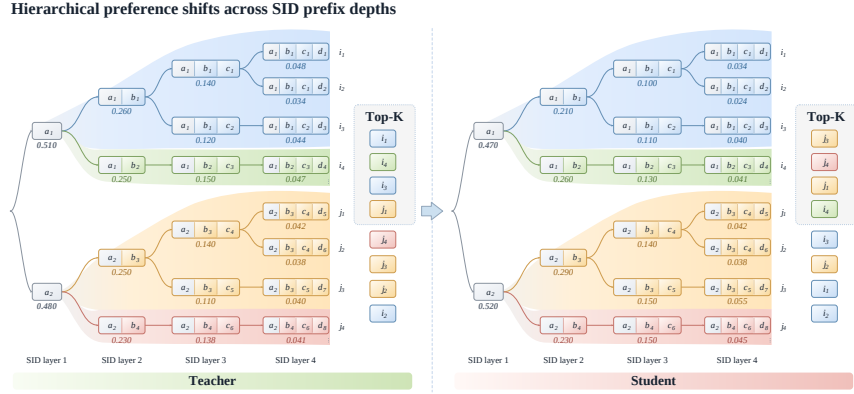


Figure 1: Hierarchical candidate-preference mismatch. Teacher and student assign different preference masses to shared SID prefix groups. The example is illustrative.

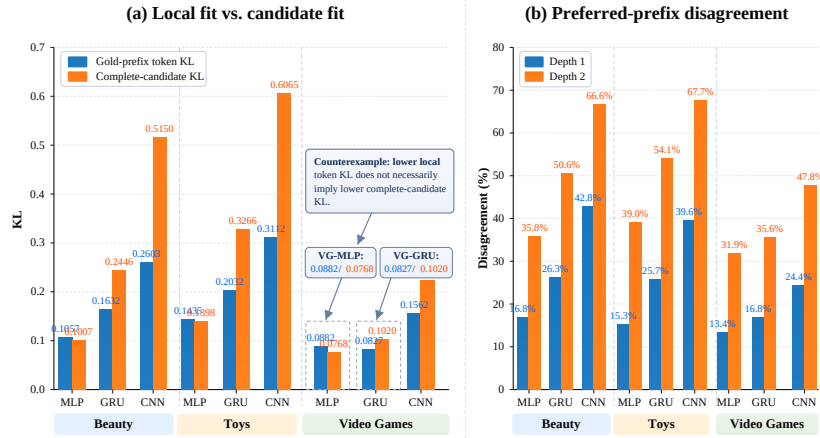


Figure 2: Candidate-level and hierarchical preference mismatches. (a) Local token agreement versus complete-candidate agreement. (b) Preferred-prefix disagreement across SID depths.

depth-wise score increments. Both objectives train the same light-weight student introduced in Section 3.1; neither adds prediction heads, alters valid-prefix constraints, or changes online beam search.

5.1 Overall Training Objective

For a training request u with observed next-item SID z^{gt} , our objective is

$$\mathcal{L}_{\text{SERVESID}} = \mathcal{L}_{\text{digit}} + \lambda_H \mathcal{L}_{\text{HServe}} + \lambda_D \mathcal{L}_{\text{DA-CDRS}}, \quad (6)$$

where $\mathcal{L}_{\text{digit}}$ is the ground-truth SID cross-entropy defined in Section 3.2, and $\lambda_H, \lambda_D \geq 0$ control the structural objectives. Unlike the baseline, SERVESID does not retain a separate gold-prefix token-KL term. Ground-truth supervision remains necessary because matching a teacher’s preferences does not guarantee relevance to the observed next item. The following losses operate on teacher-proposed complete candidates, with gradients applied only to the student. Request-level objectives are averaged across the

training minibatch, and the teacher is fixed throughout optimization, keeping the distillation targets stationary while the student adapts.

5.2 HServe: Hierarchical Serving-Distribution Alignment

Shared complete-candidate support. For each training request, we obtain the teacher’s top- M beam-reachable, valid complete SIDs and cache the resulting support $\mathcal{C}_M(u)$. Both models evaluate these same complete sequences using accumulated conditional log-probabilities $s_m(i | u)$, where $m \in \{T, S\}$. Teacher scores can be precomputed, whereas student scores are recomputed as parameters change. We define a distribution normalized within this support:

$$\pi_{m,M}(i | u) = \frac{\exp(s_m(i | u))}{\sum_{j \in \mathcal{C}_M(u)} \exp(s_m(j | u))}, \quad i \in \mathcal{C}_M(u). \quad (7)$$

This normalization makes teacher and student probabilities comparable over identical candidates. It represents preference conditional on the selected support rather than over the entire item

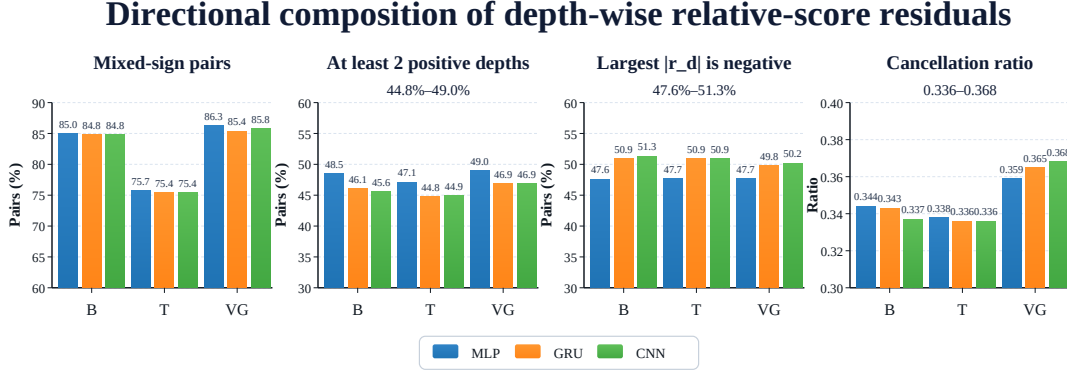


Figure 3: Directional composition of depth-wise residuals. Mixed-sign residuals, positive-residual depths, negative-dominant residuals, and cancellation ratios across datasets and student architectures.

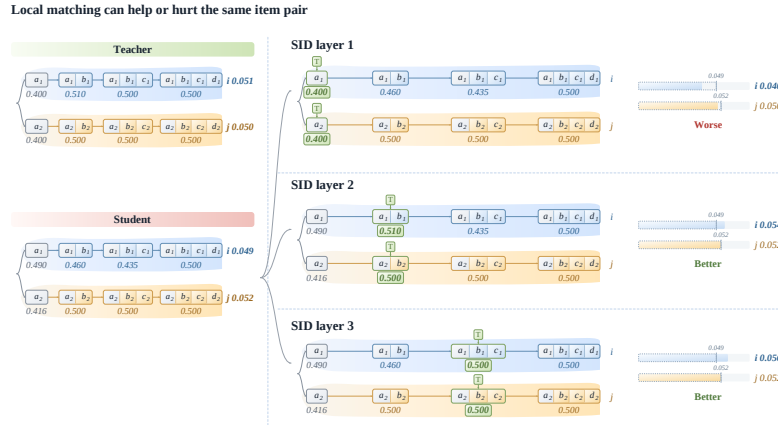


Figure 4: Directional effects of depth-wise correction. Replacing a student increment with its teacher counterpart can increase or decrease the complete-candidate score gap.

catalog, so candidates outside $\mathcal{C}_M(u)$ are not part of this KL objective; using a common support avoids interpreting differences caused merely by two models proposing different candidate lists as distributional mismatch.

Hierarchically consistent prefix marginals. As illustrated in Figure 1, shared SID prefixes form nested candidate groups whose preference mass may differ between teacher and student. A length- d prefix p defines the subset of supported candidates whose first d codes equal p . We aggregate their complete-candidate probabilities:

$$\pi_{m,M}^{(d)}(p | u) = \sum_{\substack{i \in \mathcal{C}_M(u) \\ z_1^d = p}} \pi_{m,M}(i | u). \quad (8)$$

Every candidate contributes to exactly one prefix group at each depth, so the marginals are normalized and mutually consistent across depths. At $d = L$, each distinct complete SID corresponds to

a leaf and the marginal reduces to the complete-candidate distribution. At shallower depths, it captures the total preference mass assigned to groups of candidates with common initial codes. No auxiliary prefix predictor is introduced: all marginals are deterministic aggregations of the same complete-candidate scores. These groupings are defined by code identity rather than manually assigned product categories.

Multi-level alignment. HServe minimizes the teacher-to-student KL divergence at every depth:

$$\mathcal{L}_{\text{HServe}}(u) = \sum_{d=1}^L \alpha_d \text{KL}(\pi_{T,M}^{(d)}(\cdot | u) \| \pi_{S,M}^{(d)}(\cdot | u)), \quad \alpha_d \geq 0. \quad (9)$$

The hierarchical terms impose optimization constraints at different levels of candidate aggregation. A shallow-prefix KL term penalizes discrepancies in the total preference mass assigned to competing branches, whereas the leaf-level term distinguishes individual candidates within those branches. Through the shared candidate scores, gradients from each prefix marginal propagate to

its supported descendants. Consequently, multi-depth alignment changes how the same teacher preference information is emphasized during optimization, without introducing additional prediction heads or independent supervision targets. The leaf term ($d = L$) matches complete-item preferences; earlier terms penalize discrepancies in the mass allocated to shared-prefix groups. The shallower terms contain no independent teacher information—exact leaf agreement implies agreement for every aggregated marginal—and instead impose direct multi-granularity constraints when the student cannot match the teacher exactly. Differentiation through each student marginal propagates group-level errors to the complete candidate scores that constitute it. Depth weights α_d specify the optimization emphasis; they do not change the underlying teacher distribution.

5.3 DA-CDRS: Direction-Aware Cross-Depth Relative-Score Distillation

Cross-cutoff candidate pairs. Let $\text{rank}_T(i \mid u)$ denote an item's rank within the teacher proposal list. We form teacher-ordered pairs

$$\mathcal{B}(u) = \{(i^+, i^-) : \text{rank}_T(i^+ \mid u) \leq K_{\text{tr}}, K_{\text{tr}} < \text{rank}_T(i^- \mid u) \leq B\}, \quad (10)$$

with $B \leq M$. These pairs compare teacher-preferred candidates inside a selected cutoff with candidates outside it; they are not restricted to the exact K_{tr} -th and $(K_{\text{tr}} + 1)$ -th items. The pair labels express the teacher's relative preference, not observed item relevance; incorrect teacher rankings can therefore still produce valid training pairs under this definition. Their score gaps are

$$G_m(b) = s_m(i^+ \mid u) - s_m(i^- \mid u) = \sum_{d=1}^L \Delta G_m^{(d)}(b), \quad (11)$$

where

$$\Delta G_m^{(d)}(b) = \log q_m(z_d^{i^+} \mid u, z_{<d}^{i^+}) - \log q_m(z_d^{i^-} \mid u, z_{<d}^{i^-}), \quad (12)$$

and $q_T = p_T$ by notation. These increments measure relative conditional scores, not independent depth-wise item rankings.

Direction-aware depth weighting. Define the teacher-student increment residual

$$r_d(b) = \Delta G_T^{(d)}(b) - \Delta G_S^{(d)}(b). \quad (13)$$

If two candidates share the same prefix before depth $d_{\text{div}}(b)$, their relative increments cancel at earlier depths. We therefore use the validity mask

$$v_d(b) = \mathbf{1}[d \geq d_{\text{div}}(b)]. \quad (14)$$

Figure 4 illustrates that replacing a student increment with its teacher counterpart can move the complete score gap in either direction. For each eligible depth, we hypothetically replace only the student's relative-score increment with the teacher's, giving

$$\tilde{G}_{S,d}(b) = G_S(b) + r_d(b). \quad (15)$$

We score how much each replacement moves the complete gap toward the teacher-preferred direction:

$$A_d(b) = v_d(b) [\tilde{G}_{S,d}(b) - G_S(b)]_+ = v_d(b) [r_d(b)]_+, \quad (16)$$

and normalize these scores into detached weights,

$$w_d(b) = \text{stopgrad} \left(\frac{A_d(b)}{\sum_{t=1}^L A_t(b) + \epsilon} \right). \quad (17)$$

A depth thus receives weight only when its teacher increment moves the complete score gap in the teacher-preferred direction, and the weight grows with the magnitude of that favorable residual. Because $A_d(b)$ reduces to the positive part of the increment residual, this attribution is an operational positive-residual weighting over the complete gap rather than a causal estimate of depth importance after retraining; the “decision” it references is the direction of the complete candidate score gap. If every eligible residual in a pair is nonpositive, the pair contributes zero DA loss. The weights are detached during backpropagation but recomputed as student scores evolve, allowing the allocation of supervision to adapt throughout training.

Scale-normalized correction. To make residual matching less sensitive to depth-dependent score scales, let

$$\sigma_d = \max \left(\text{Std}_{\text{train,valid}}[\Delta G_T^{(d)}], 10^{-4} \right). \quad (18)$$

We use a Huber penalty ρ_H for the signed relative-score difference:

$$\ell_{\text{DA}}(b) = \lambda_{\text{corr}} \sum_{d=1}^L w_d(b) \rho_H \left(\frac{\Delta G_S^{(d)}(b) - \Delta G_T^{(d)}(b)}{\sigma_d} \right). \quad (19)$$

The request-level loss averages ℓ_{DA} over $\mathcal{B}(u)$, using zero contribution when no pair is available. Attribution-weighted increment matching thus determines which local teacher differences receive stronger explicit supervision. Since $A_d(b) = v_d(b)[r_d(b)]_+$, the weighting is equivalent to normalized positive-residual weighting over eligible depths; its value relative to uniform weighting is examined empirically in Section 6.3.

5.4 Offline Training and Verify-Free Inference

We construct teacher proposal lists, candidate scores, pair relations, and teacher increment statistics from training requests only. The frozen teacher's data are cached; student scores, marginals, replacements, and attribution weights are computed during optimization. No validation or test request contributes teacher targets to the training loss.

At deployment, all teacher-dependent caches and distillation computations are discarded. The student retains one history-encoding pass, depth-wise SID prediction, validity masking, and constrained beam expansion. Consequently, the deployed student retains the original verify-free inference graph without introducing teacher-dependent scoring, auxiliary prediction heads, or reranking stages. The proposed objectives modify only the offline training procedure and do not constitute a new decoding algorithm. Offline cache generation, storage, and additional student scoring nevertheless increase training-side cost; they should be reported separately from serving latency. The method targets recommendation quality under an unchanged verify-free inference graph, not a guaranteed preservation of every teacher Top- K item.

6 Experiments

We evaluate SERVESID along three questions: (RQ1) Does structure-aware distillation improve next-item recommendation across datasets and student architectures? (RQ2) Does HServe add value beyond complete-candidate, leaf-only distribution matching? (RQ3) Does direction-aware depth-wise distillation complement hierarchical alignment and outperform uniform depth weighting?

6.1 Experimental Setup

Datasets and evaluation. We use Amazon Beauty and Toys from Amazon Reviews 2014 and Video Games from Amazon Reviews 2023. User interactions are ordered chronologically, with leave-one-out next-item evaluation and at most 20 recent interactions retained as request history. All methods within a dataset share identical preprocessing, splits, SID assignments, and validity constraints. We report Recall@5/10 and NDCG@5/10 against the held-out next item. Recall@ K measures whether the held-out next item appears among the top- K recommendations, while NDCG@ K additionally accounts for its ranking position. These relevance-based metrics assess recommendation utility independently of teacher-student preference agreement.

Teacher and students. A frozen TIGER teacher provides supervision in an RQ-VAE SID space with four codes per item and codebook size 256. We distill the teacher into three lightweight students: an MLP (1.664M trainable parameters), a GRU with hidden size 384 (1.201M), and a CNN with embedding dimension 256 (1.753M). These are distinct architectures, not an ordered capacity series. Students share the teacher’s SID inventory and employ constrained beam search with width 50; the teacher’s beam-20 results are reference measurements rather than matched-cost comparisons. Fixing the SID inventory and validity constraints ensures that differences among training objectives are not attributable to changes in item tokenization or admissible candidate sequences. Student beam width is held constant within each architecture.

Training configuration. For every training request, the offline teacher cache contains up to $M = 64$ valid complete SIDs and their accumulated scores. HServe aligns distributions on this shared support with depth weights $\alpha_d = 1$ and $\lambda_H = 4$. DA-CDRS constructs teacher-ordered cross-cutoff pairs between ranks 1–10 and 11–40 for weighted depth-wise relative-score distillation, with $\lambda_D = 1$ and increment-correction weight $\lambda_{\text{corr}} = 31.42$; the uniform variant replaces attribution weights with equal weights over eligible SID depths. Models are trained with learning rate 5×10^{-5} and seeds 42, 43, 44. All teacher scores, pairs, and increment-scale statistics originate from training requests only; validation and test teacher outputs are used exclusively for independent diagnostics. The base KD objective combines digit CE with gold-prefix token KL, whereas the SERVESID objective combines digit CE with HServe and DA-CDRS without an additional token-KL term. All methods follow the same validation-based checkpoint selection protocol.

Baselines. We compare against representative distillation objectives: RD and RRD transfer ranking structure; GKD and DistILLM-2 provide generation-aware distillation; RCE-KD adapts cross-entropy distillation to ranking objectives; SmartGR and LOHRec exploit SID- or identifier-specific signals. Each method keeps its core supervision principle under the same teacher, tokenizer, backbone,

and protocol; adaptations are detailed in the supplementary material. All baseline adaptations use the same next-item evaluation protocol and validation-based checkpoint selection rule. This controls for differences in student architecture and evaluation setup when comparing supervision objectives derived from ranking-based, generation-based, and SID-specific distillation methods.

6.2 Overall Recommendation Performance

Table 1 compares SERVESID with conventional recommendation distillation, general generative distillation, and SID-/identifier-aware methods across three datasets and three student architectures. We evaluate all methods using held-out Recall and NDCG under the same student architectures and decoding configurations. **[TODO: After the no-pair-loss checkpoints are verified, report here (i) the NDCG@10 advantage across the nine dataset-student settings, (ii) the gains relative to the corresponding base-KD student, and (iii) whether MLP, GRU, and CNN exhibit different benefit patterns.]**

6.3 Ablation Study

Figures 1 and 2 illustrate and quantify hierarchical candidate-preference mismatches, while Figures 3 and 4 characterize the directional structure and correction effects of depth-wise relative-score residuals. These observations motivate the complementary HServe and DA-CDRS objectives. Table 2 examines the individual and complementary contributions of HServe and DA-CDRS. Comparing HServe-only with d4-only KL evaluates hierarchical alignment beyond leaf-level matching. DA-CDRS-only isolates depth-wise relative-score supervision, while the uniform-weight variant tests the role of directional weighting. Comparisons with the full model assess the complementarity of the two objectives. **[TODO: After the no-pair-loss checkpoints are verified, add the ablation conclusions here (HServe vs. d4-only, direction-weighted vs. uniform, and complementarity).]**

6.4 Training Cost and Scope

SERVESID changes only the offline training objective. The deployed student retains the same history encoder, SID prediction heads, validity masking, and constrained beam search as its baseline counterpart, with no teacher queries or additional reranker. Therefore, the proposed distillation losses do not introduce extra serving-time model components. Any throughput or latency advantage relative to the autoregressive TIGER teacher is primarily attributable to the lightweight student architecture rather than to a newly accelerated decoder.

The efficiency trade-off is instead concentrated offline. Teacher candidate generation and caching, evaluation of complete candidates during student training, construction of cross-cutoff pairs, and calculation of depth attribution all add training-side computation or storage. These costs should be distinguished from online inference costs, and numerical claims about identical latency or overall efficiency require matched-hardware measurements. Our empirical scope is further restricted to one frozen TIGER teacher, one SID tokenizer, and three Amazon categories. HServe aligns preferences only within teacher-proposed supports, while DA-CDRS

Dataset	Category	Method	MLP				GRU				CNN			
			R@5	N@5	R@10	N@10	R@5	N@5	R@10	N@10	R@5	N@5	R@10	N@10
	Traditional	RD	0.0422	0.0281	0.0657	0.0357	0.0433	0.0285	0.0671	0.0362	0.0419	0.0274	0.0649	0.0348
		RDR	0.0430	0.0287	0.0659	0.0361	0.0422	0.0281	0.0664	0.0359	0.0420	0.0281	0.0649	0.0354
	General	RCE-KD	0.0434	0.0286	0.0666	0.0363	0.0432	0.0286	0.0671	0.0361	0.0409	0.0276	0.0651	0.0354
		GKD	0.0428	0.0286	0.0668	0.0363	0.0416	0.0281	0.0656	0.0358	0.0380	0.0254	0.0580	0.0319
		DistiLLM-2	0.0428	0.0287	0.0666	0.0361	0.0416	0.0281	0.0656	0.0358	0.0380	0.0254	0.0580	0.0319
		SID-MLP (base KD)	0.0429	0.0282	0.0660	0.0356	0.0416	0.0281	0.0656	0.0358	0.0380	0.0254	0.0580	0.0319
		SmartGR	0.0434	0.0286	0.0668	0.0363	0.0426	0.0286	0.0669	0.0361	0.0374	0.0250	0.0594	0.0320
		LOHRec	0.0414	0.0278	0.0639	0.0350	0.0376	0.0251	0.0576	0.0315	0.0361	0.0242	0.0542	0.0300
	SERVESID		0.0442	0.0292	0.0679	0.0369	0.0441	0.0291	0.0682	0.0368	0.0437	0.0288	0.0686	0.0368
	Traditional	RD	0.0380	0.0234	0.0615	0.0309	0.0381	0.0245	0.0614	0.0318	0.0380	0.0241	0.0603	0.0312
		RDR	0.0396	0.0245	0.0625	0.0319	0.0381	0.0245	0.0622	0.0322	0.0378	0.0239	0.0585	0.0306
	General	RCE-KD	0.0387	0.0244	0.0631	0.0324	0.0380	0.0246	0.0622	0.0324	0.0364	0.0232	0.0587	0.0303
		GKD	0.0373	0.0237	0.0627	0.0318	0.0382	0.0243	0.0613	0.0316	0.0328	0.0210	0.0533	0.0276
		DistiLLM-2	0.0375	0.0238	0.0614	0.0315	0.0367	0.0233	0.0565	0.0296	0.0328	0.0210	0.0533	0.0276
		SID-MLP (base KD)	0.0374	0.0235	0.0626	0.0315	0.0367	0.0233	0.0565	0.0296	0.0328	0.0210	0.0533	0.0276
		SmartGR	0.0381	0.0241	0.0623	0.0319	0.0382	0.0242	0.0604	0.0314	0.0329	0.0206	0.0520	0.0267
		LOHRec	0.0362	0.0224	0.0596	0.0299	0.0317	0.0201	0.0510	0.0263	0.0320	0.0198	0.0514	0.0260
	SERVESID		0.0408	0.0254	0.0644	0.0330	0.0388	0.0250	0.0650	0.0334	0.0398	0.0250	0.0639	0.0328
	Traditional	RD	0.0575	0.0372	0.0927	0.0485	0.0595	0.0388	0.0938	0.0498	0.0580	0.0373	0.0921	0.0483
		RDR	0.0592	0.0390	0.0918	0.0496	0.0596	0.0393	0.0892	0.0490	0.0591	0.0389	0.0884	0.0483
	General	RCE-KD	0.0593	0.0388	0.0926	0.0497	0.0595	0.0394	0.0938	0.0503	0.0589	0.0387	0.0919	0.0493
		GKD	0.0593	0.0388	0.0929	0.0496	0.0597	0.0393	0.0937	0.0502	0.0569	0.0370	0.0896	0.0475
		DistiLLM-2	0.0594	0.0387	0.0927	0.0494	0.0596	0.0392	0.0939	0.0503	0.0587	0.0384	0.0914	0.0489
		SID-MLP (base KD)	0.0588	0.0383	0.0922	0.0488	0.0591	0.0389	0.0935	0.0499	0.0553	0.0358	0.0870	0.0460
		SmartGR	0.0593	0.0388	0.0929	0.0496	0.0594	0.0391	0.0933	0.0500	0.0549	0.0359	0.0870	0.0462
		LOHRec	0.0584	0.0377	0.0908	0.0481	0.0538	0.0346	0.0818	0.0436	0.0464	0.0297	0.0711	0.0376
	SERVESID		0.0604	0.0396	0.0944	0.0505	0.0607	0.0400	0.0954	0.0512	0.0605	0.0399	0.0944	0.0508

Table 1: Overall Top- K recommendation performance. Methods are grouped by family across three datasets and three student architectures. Bold highlights the best result in each column; gray marks second-best results (ties included).

Arm	MLP				GRU				CNN			
	R@5	N@5	R@10	N@10	R@5	N@5	R@10	N@10	R@5	N@5	R@10	N@10
<i>Beauty</i>												
SID-MLP (base KD)	0.0429	0.0282	0.0660	0.0356	0.0416	0.0281	0.0656	0.0358	0.0380	0.0254	0.0580	0.0319
d4-only KL	0.0431	0.0283	0.0663	0.0359	0.0422	0.0283	0.0660	0.0360	0.0402	0.0266	0.0625	0.0348
DA-CDRS only	0.0435	0.0286	0.0672	0.0362	0.0430	0.0286	0.0673	0.0364	0.0414	0.0273	0.0652	0.0359
HServe only	0.0436	0.0287	0.0670	0.0363	0.0431	0.0286	0.0671	0.0364	0.0418	0.0276	0.0656	0.0360
HServe + Uniform	0.0437	0.0286	0.0673	0.0364	0.0437	0.0287	0.0674	0.0365	0.0423	0.0280	0.0669	0.0362
SERVESID(full)	0.0442	0.0292	0.0679	0.0369	0.0441	0.0291	0.0682	0.0368	0.0437	0.0288	0.0686	0.0368
<i>Toys</i>												
SID-MLP (base KD)	0.0374	0.0235	0.0626	0.0315	0.0367	0.0233	0.0565	0.0296	0.0328	0.0210	0.0533	0.0276
d4-only KL	0.0379	0.0237	0.0628	0.0317	0.0369	0.0237	0.0594	0.0313	0.0354	0.0225	0.0566	0.0305
DA-CDRS only	0.0389	0.0246	0.0638	0.0324	0.0377	0.0241	0.0622	0.0323	0.0370	0.0236	0.0598	0.0322
HServe only	0.0390	0.0245	0.0633	0.0323	0.0379	0.0243	0.0630	0.0325	0.0375	0.0238	0.0606	0.0320
HServe + Uniform	0.0395	0.0248	0.0640	0.0325	0.0381	0.0247	0.0642	0.0328	0.0382	0.0242	0.0630	0.0323
SERVESID(full)	0.0408	0.0254	0.0644	0.0330	0.0388	0.0250	0.0650	0.0334	0.0398	0.0250	0.0639	0.0328
<i>VG</i>												
SID-MLP (base KD)	0.0588	0.0383	0.0922	0.0488	0.0591	0.0389	0.0935	0.0499	0.0553	0.0358	0.0870	0.0460
d4-only KL	0.0589	0.0384	0.0925	0.0491	0.0593	0.0390	0.0937	0.0502	0.0570	0.0370	0.0888	0.0488
DA-CDRS only	0.0595	0.0388	0.0932	0.0498	0.0601	0.0394	0.0943	0.0507	0.0586	0.0380	0.0914	0.0498
HServe only	0.0597	0.0390	0.0935	0.0499	0.0600	0.0395	0.0945	0.0506	0.0591	0.0386	0.0923	0.0500
HServe + Uniform	0.0599	0.0391	0.0938	0.0500	0.0602	0.0397	0.0950	0.0510	0.0596	0.0390	0.0935	0.0502
SERVESID(full)	0.0604	0.0396	0.0944	0.0505	0.0607	0.0400	0.0954	0.0512	0.0605	0.0399	0.0944	0.0508

Table 2: Ablation of HServe and DA-CDRS. Structural variants compare hierarchical distribution alignment, depth-wise supervision, and their combination.

supplies an optimization-based rather than causal depth attribution.

7 Conclusion and Limitations

We investigate structured candidate-preference distillation for light-weight SID recommendation. SERVESID exploits two complementary structures of complete-candidate preferences: hierarchical prefix-group organization and depth-wise relative-score decomposition. HServe aligns preference distributions across SID granularities, while DA-CDRS allocates supervision to selected relative-score increments.

Both objectives jointly train a single lightweight student without modifying its verify-free inference architecture.

Our evaluation covers one teacher family, one SID tokenizer, and three product categories. Teacher-proposed supports may omit relevant items, and the positive-residual weighting is a detached operational rule that does not explicitly consider the complete student score gap and is not a causal measure of depth importance. Broader teacher, tokenizer, and deployment settings remain important directions for future study.

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